ETH zürich

Information Retrieval #2 - Boolean Retrieval

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Inhaltsverzeichnis/Agenda

1. Lecture Recap
2. Exercise 1: Boolean Retrieval
3. Exam Questions
4. (Kahoot)

Lecture this week: Boolean Retrieval

- Grepping
  - Shortcomings?
- Term, Document
  - Inclusion, Occurrence, Order
  - Set vs. Bag vs. List
- Incidence Matrix
  - Querying on Incidence Matrices
- Improving the Incidence Matrix
  - Inverted Index
  - Intersection & Union Algorithm
- Evaluating Results
  - Precision, Recall

Grepping

- “grep lawyer corpus.txt”
  - What does this do?

- What shortcomings does grepping have?
  - You need to download the entire web
  - No capability of proximity search
  - No capability of ranking results

```
$ grep lawyer corpus.txt

    William Whyte was. Some pragmatical seventeenth century lawyer, I
    He was a lawyer. That sounded ominous. What was the relation between
    "Her banker or her lawyer. There is that double possibility. But I am
    "To an English lawyer named Norton."
    and send down to Fordham, the Horsham lawyer.'
    "I did as he ordered, and when the lawyer arrived I was asked to step
    "I signed the paper as directed, and the lawyer took it away with
    violin-player, boxer, swordsman, lawyer, and self-poisoner by cocaine
    which was a Penang-lawyer weighted with lead, was just such a weapon
    a lawyer with a good practice. They had one child, but the yellow
    has some claim on half Cunningham's estate, and the lawyers have been
    lawyer whose name was given in the paper. There we met two gentlemen,
    as he followed the argument of the lawyer. At least, I thought, when
    is known as a "Penang lawyer." Just under the head was a broad silver
    Holmes to the porter. "A lawyer, is he not, gray-headed, and walks
    Reilly the lawyer and take the defense upon myself. Take my word for
    "Exactly," said I. "A plausible lawyer could make it out as an act of
    daughter of Augusto Barelli, who was the chief lawyer and once the
    I showed it to Mr. Sutro, my lawyer, who lives in Harrow. He said to
    lawyer of yours a capable man?"
    This Sutro, of course, is her lawyer. I made a mistake, I fear, in
    gentleman, who introduced himself as the lawyer, together with a
    "I only heard of it this morning," the lawyer explained.
    I have been recommended to you by my lawyers, but indeed the matter
    to the American lawyer. ... Very good. Good-bye!"
    lawyer burst excitedly into the room.
    "I can only suppose that this American lawyer put it in himself. What
    say nothing, as her lawyer had advised her to reserve her defence. We
```

Terminology

- Term: Sherlock
  lawyer
  Switzerland
  Unterwalden nid dem Wald
  ETH Zürich
  person
  watch
  run
  paper
  book
  ...

- Document: contains many terms

- Inclusion: A term is **included** in a document.
  - “Hölzli” ∈ D
- Occurrence: A term can **occur** x times in a document
- Order: A term can occur **before** another

I han es Zündhölzli azündt

Und das het e Flamme gäh

Und i ha für d'Zigarette

Welle Füür vom **Hölzli** näh

Aber ds Hölzli isch dervo-

Gspickt und uf e Teppich cho

Und es hätt no fasch es Loch i Teppich gäh dervo

Terminology

- Term: Sherlock
  lawyer
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  ETH Zürich
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  watch
  run
  paper
  book
  ...

- Document: contains many terms

- Inclusion: A term is **included** in a document.
- Occurrence: A term can **occur** x times in a document
  - #(“Hölzli, D) = 2
- Order: A term can occur **before** another

I han es Zündhölzli azündt

Und das het e Flamme gäh

Und i ha für d'Zigarette

Welle Füür vom **Hölzli** näh

Aber ds **Hölzli** isch dervo-

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Terminology

- Term: Sherlock
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- Document: contains many terms

- Inclusion: A term is **included** in a document.
- Occurrence: A term can **occur** x times in a document
- Order: A term can occur **before** another
  - “Flamme” < “Füür”

I han es Zündhölzli azündt

Und das het e **Flamme** gäh

Und i ha für d Zigarette

Welle **Füür** vom Hölzli näh

Aber ds Hölzli isch dervo-

Gspickt und uf e Teppich cho

Und es hätt no fasch es Loch i Teppich gäh dervo

Set vs. Bag vs. List

- Which of the three is this? Inclusion, Occurrence, Order?

I han es Zündhölzli azündt

Und das het e Flamme gäh

Und i ha für d'Zigarette

Welle Füür vom Hölzli näh

Aber ds Hölzli isch dervo-

Gspickt und uf e Teppich cho

Und es hätt no fasch es Loch i Teppich gäh dervo

Set vs. Bag vs. List

This is a list. Lists preserve Inclusion, Occurrence, Order

I han es Zündhölzli azündt

Und das het e Flamme gäh

Und i ha für d'Zigarette

Welle Füür vom Hölzli näh

Aber ds Hölzli isch dervo-

Gspickt und uf e Teppich cho

Und es hätt no fasch es Loch i Teppich gäh dervo

Set vs. Bag vs. List

- Which of the three is this? Inclusion, Occurrence, Order?

Set vs. Bag vs. List

This is a bag. Bags preserve Inclusion, Occurrence, Order

Set vs. Bag vs. List

- Which of the three is this? Inclusion, Occurrence, Order?

Set vs. Bag vs. List

This is a set. Sets preserve Inclusion, Occurrence, Order

Incidence Matrix

$$\text{contains}_{i,j} = \begin{cases} 1 & \text{if document j contains term i} \\ 0 & \text{if document j does not contain term i} \end{cases}$$

Query: u

$$\begin{array}{cc} & \text{Documents} \\ & \begin{array}{cccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \end{array} \\ \text{Terms} \begin{array}{c} t \\ u \\ v \\ w \\ x \\ y \end{array} & \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & \boxed{1} & 0 & 0 & 1 & 0 & 1 \end{pmatrix} \end{array}$$

$$\text{contains}_{y,5} = 1$$

Incidence Matrix

Query: u

Documents

Terms	1	2	3	4	5	6	7	8	9	10
t	1	0		1					1	1
u	0	0	1	0	1	1	1	1	0	0
v	0	1	1	1	0	1	0	1	0	1
w	0	0	0	1	1	0	0	0	0	0
x	1	0	1	1	1	0	1	0	0	1
y	0	0	0	0	1	0	0	1	0	1

Incidence Matrix

$$\text{contains}_{i,j} = \begin{cases} 1 & \text{if document j contains term i} \\ 0 & \text{if document j does not contain term i} \end{cases}$$

Query: NOT u

$$\begin{array}{cc} & \text{Documents} \\ & \begin{array}{cccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \end{array} \\ \text{Terms} \begin{array}{c} t \\ u \\ v \\ w \\ x \\ y \end{array} & \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & \boxed{1} & 0 & 0 & 1 & 0 & 1 \end{pmatrix} \end{array}$$

$$\text{contains}_{y,5} = 1$$

Incidence Matrix

Query: NOT u

Documents

	1	2	3	4	5	6	7	8	9	10
u	0	0	1	0	1	1	1	1	0	0
NOT u	1	1	0	1	0	0	0	0	1	1

NOT

Incidence Matrix

$$\text{contains}_{i,j} = \begin{cases} 1 & \text{if document j contains term i} \\ 0 & \text{if document j does not contain term i} \end{cases}$$

Query: u AND v

$$\text{Terms} \quad \begin{array}{c} \\ t \\ u \\ v \\ w \\ x \\ y \end{array} \overset{\text{Documents}}{\begin{array}{c} \begin{array}{cccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \end{array} \\ \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & \boxed{1} & 0 & 0 & 1 & 0 & 1 \end{pmatrix} \end{array}}$$

$$\text{contains}_{y,5} = 1$$

Incidence Matrix

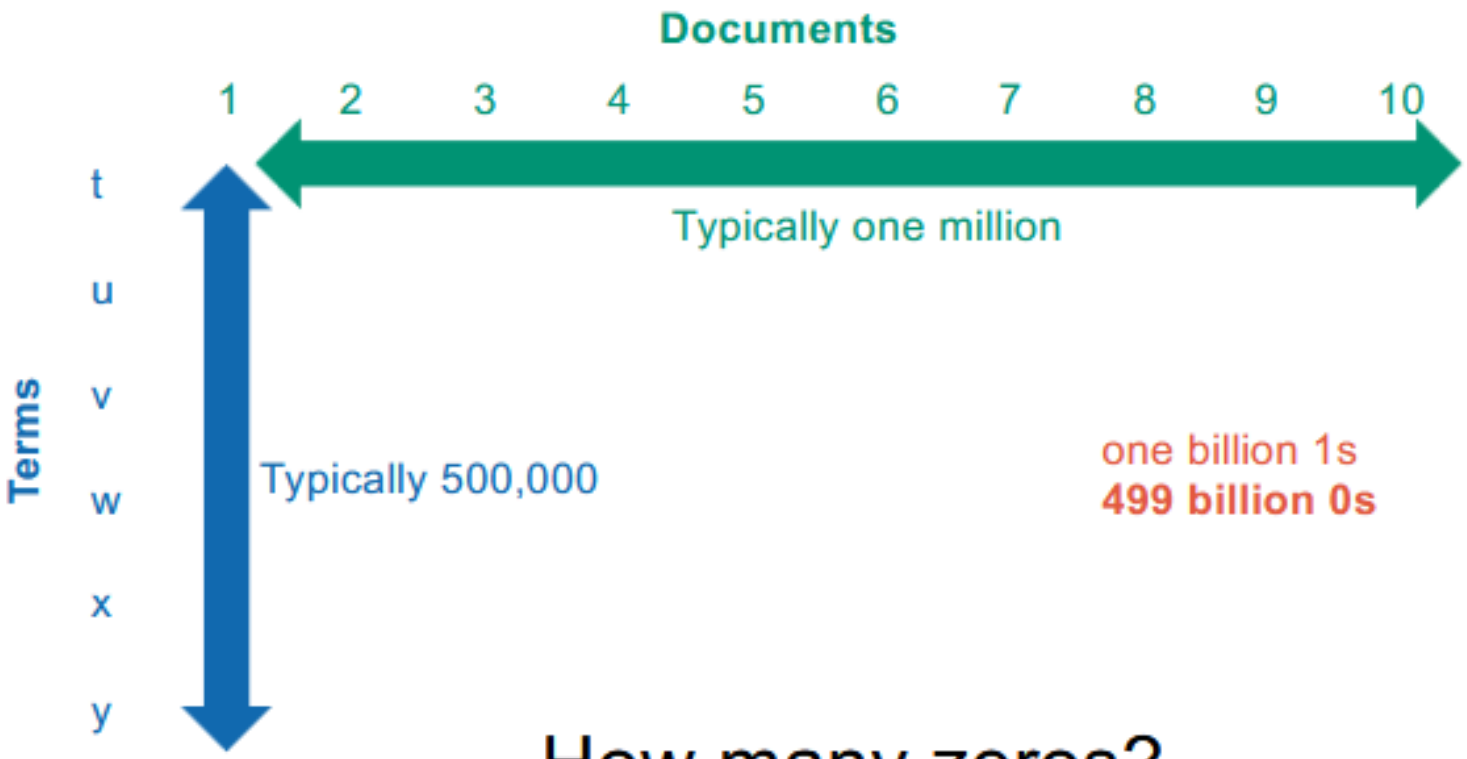
$$\text{contains}_{i,j} = \begin{cases} 1 & \text{if document j contains term i} \\ 0 & \text{if document j does not contain term i} \end{cases}$$

Query: u OR v

$$\begin{array}{cc} & \text{Documents} \\ & \begin{array}{cccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \end{array} \\ \text{Terms} \begin{array}{c} t \\ u \\ v \\ w \\ x \\ y \end{array} & \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & \boxed{1} & 0 & 0 & 1 & 0 & 1 \end{pmatrix} \end{array}$$

$$\text{contains}_{y,5} = 1$$

Incidence Matrix



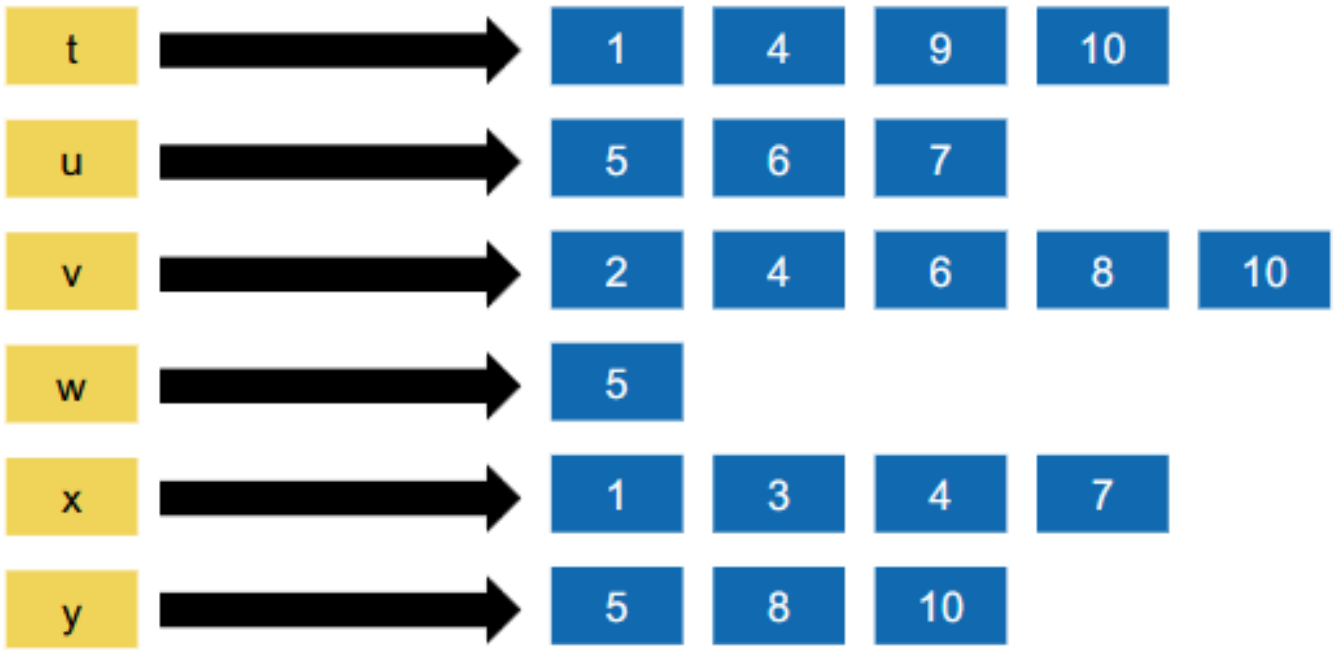
one billion 1s
499 billion 0s

How many zeros?

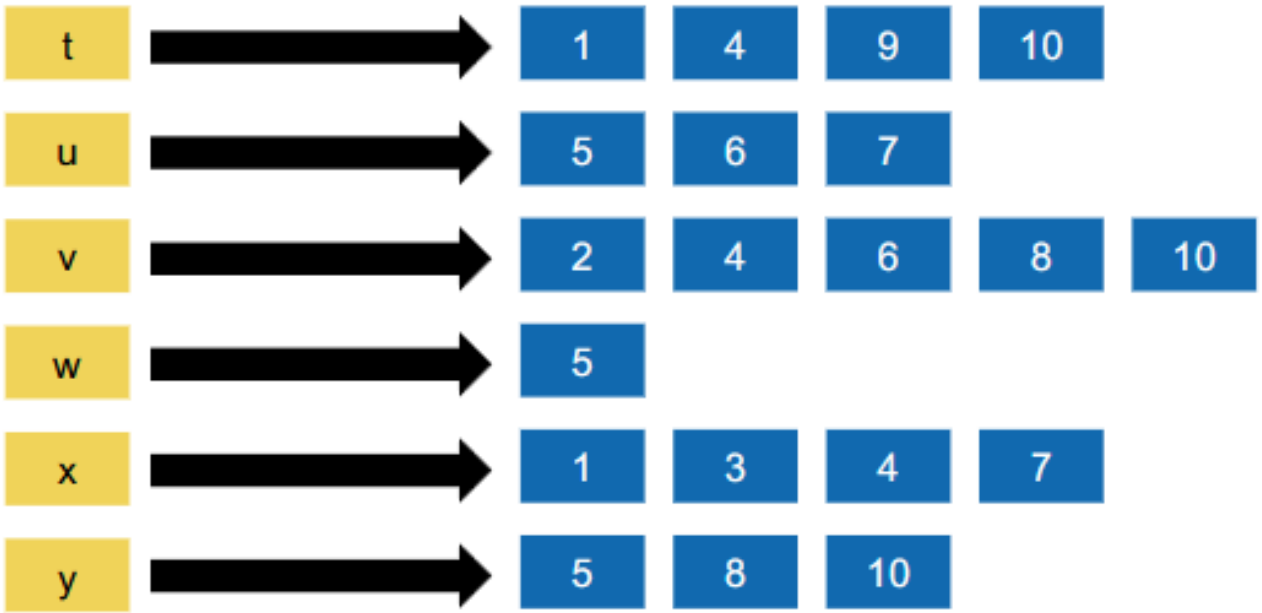
Inverted Index

Terms	Documents 1	2	3	4	5	6	7	8	9	10
t	✓	1, 4, 9, 10		✓					✓	✓
u		5, 6, 7			✓	✓	✓			
v		✓	2, 4, 6, 8, 10	✓		✓		✓		✓
w	5				✓					
x	✓	1, 3, 4, 7	✓	✓			✓			
y	5, 8, 10				✓			✓		✓

Inverted Index



Inverted Index



Term: t, u, v, …

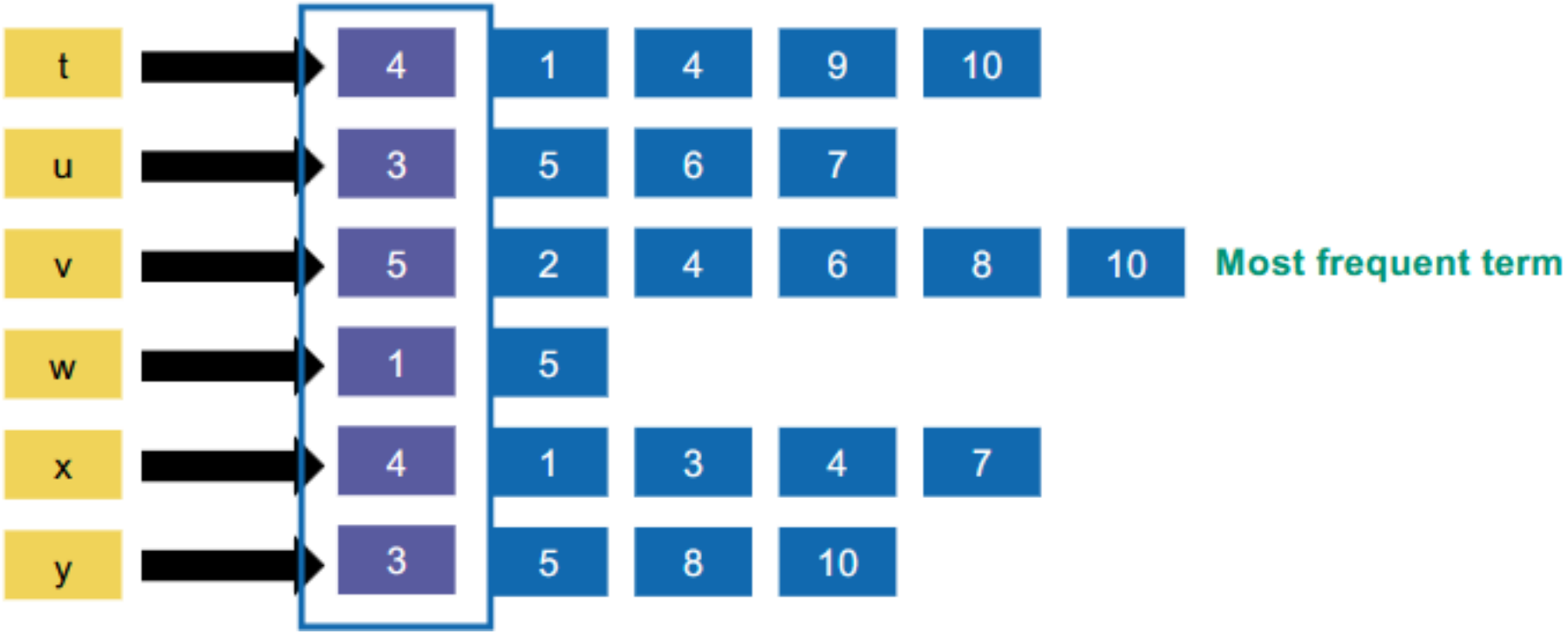
Dictionary: all terms

Document: all occurrences of its docID (e.g. doc 4 is t, v, x)

Posting: one entry on RHS (e.g. (x, 4)

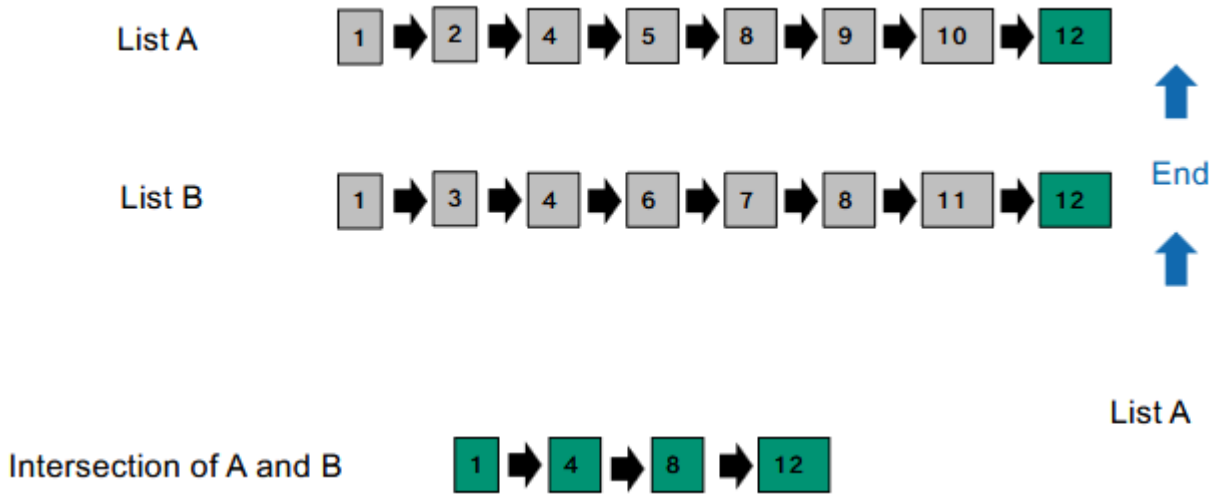
Postings list: entire RHS of a term

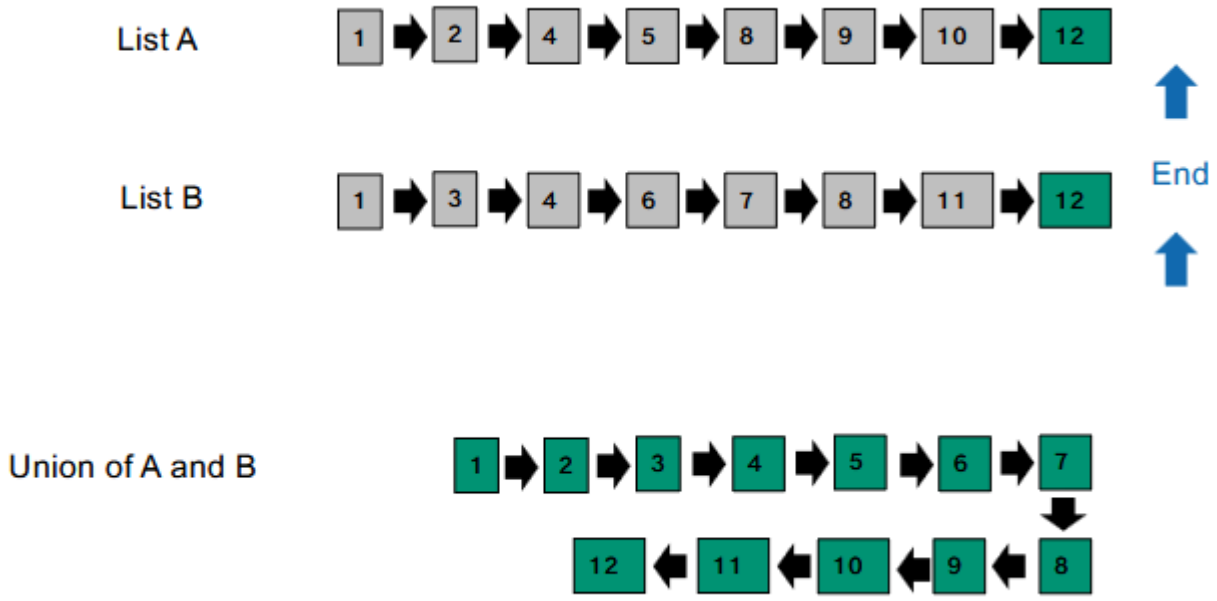
Inverted Index



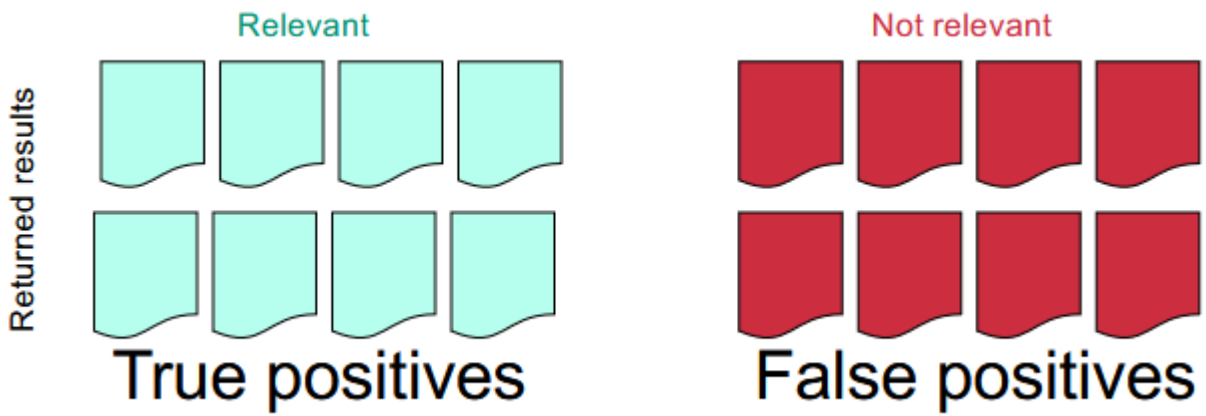
Document frequencies!

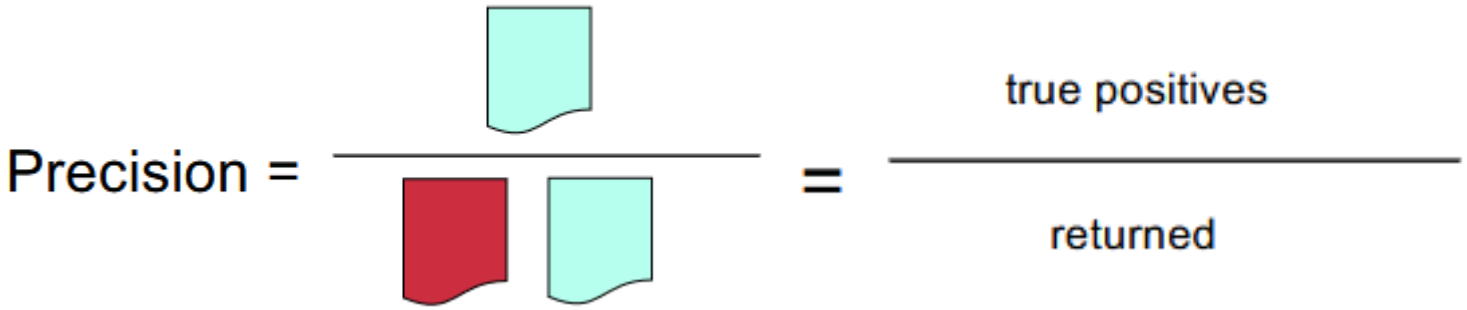
Inverted Index



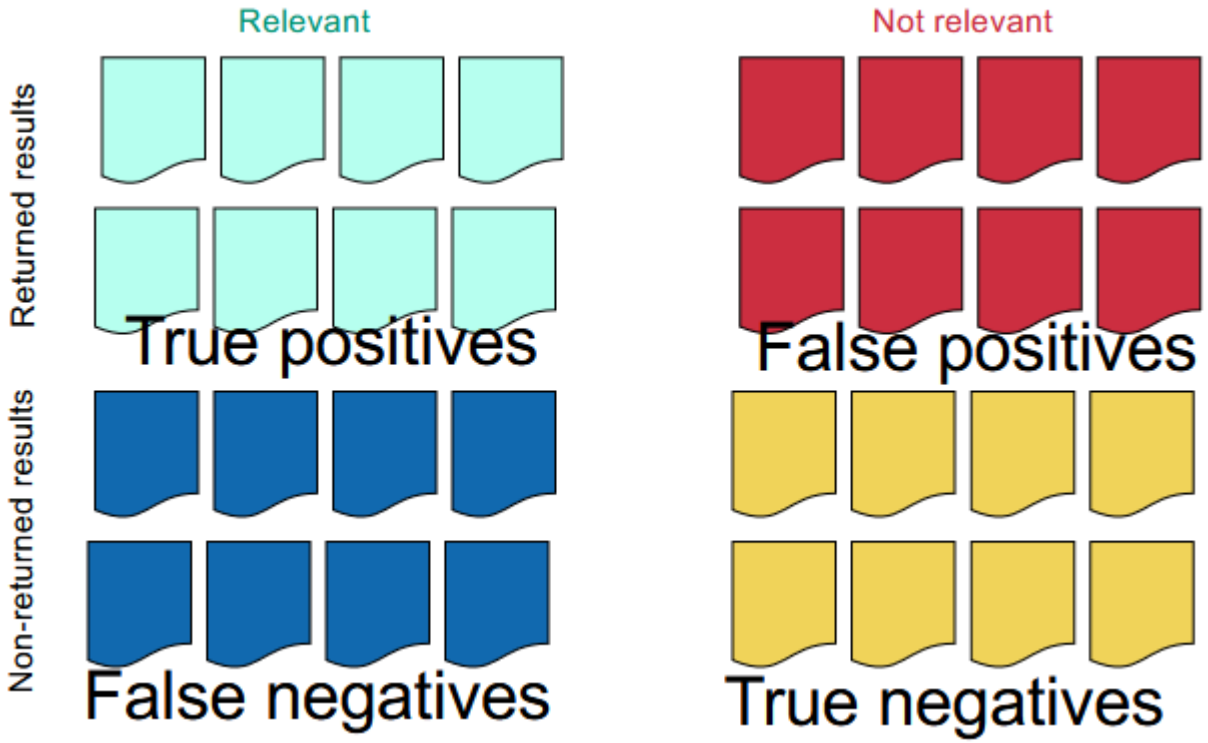


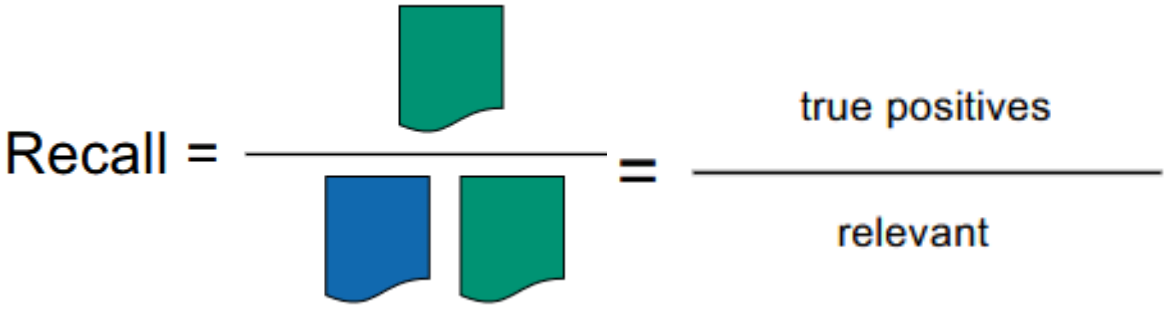
Evaluating our Results



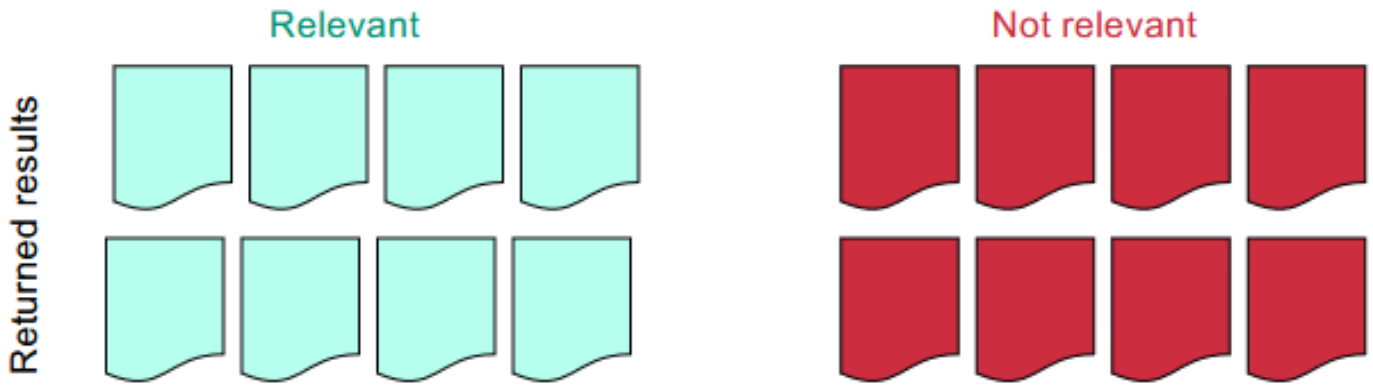


Evaluating our Results



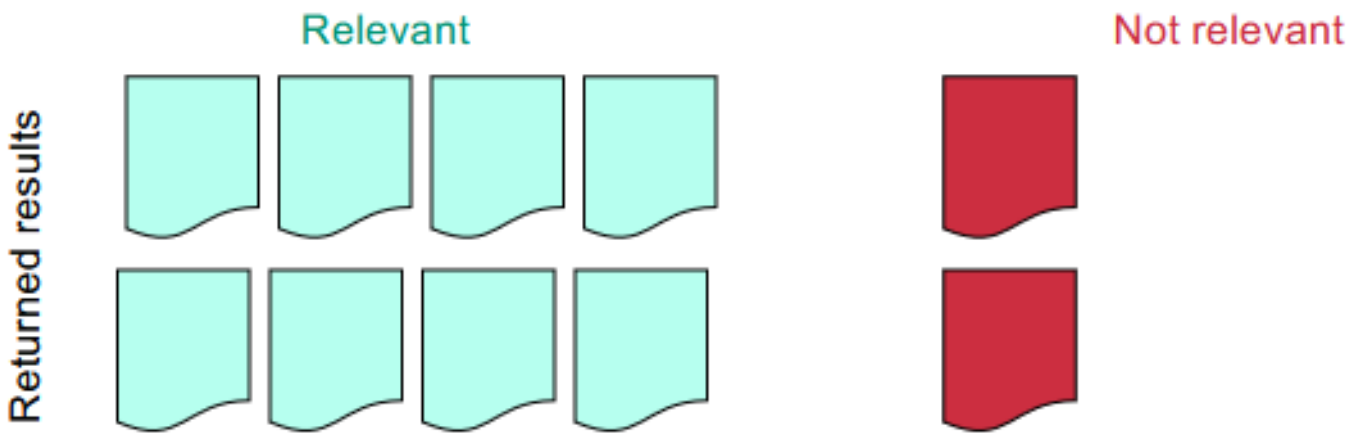


Evaluating our Results



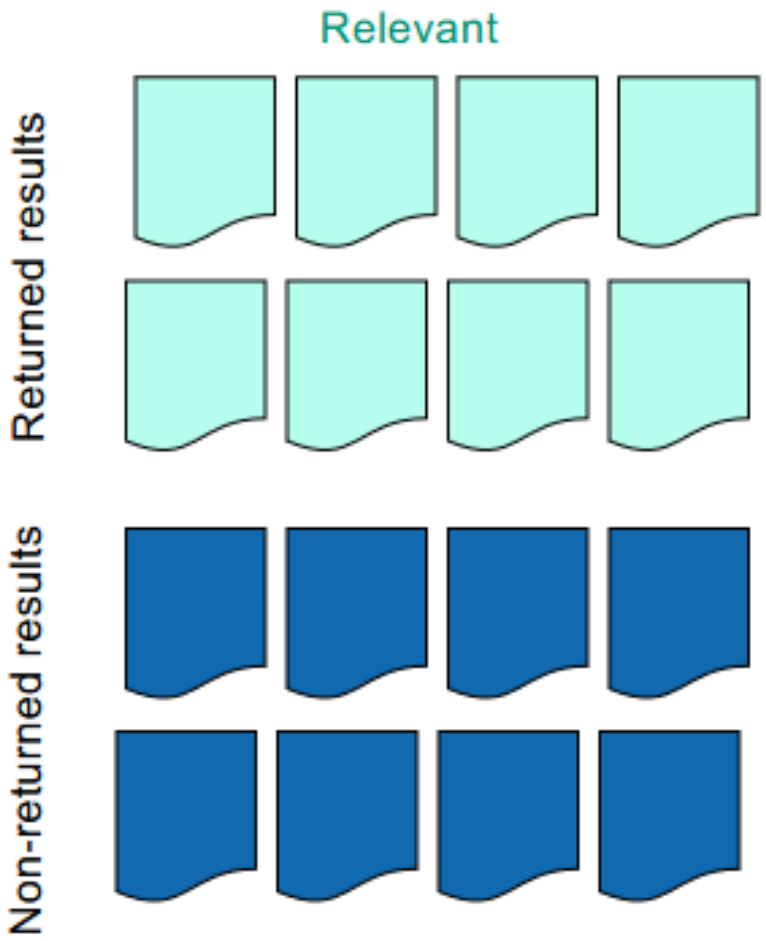
Precision: 50%

Evaluating our Results



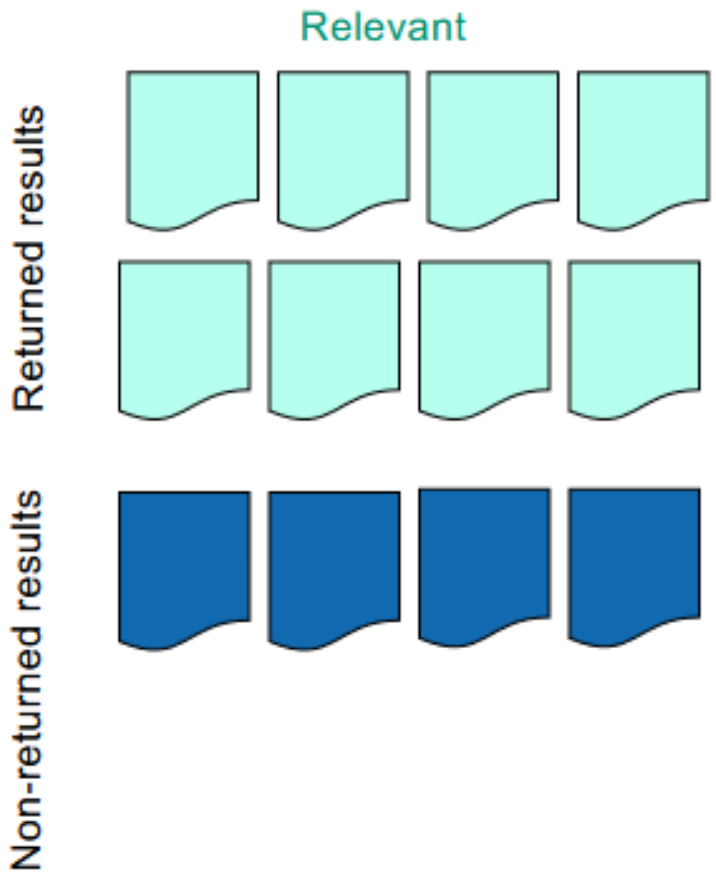
Precision: 80%

Evaluating our Results



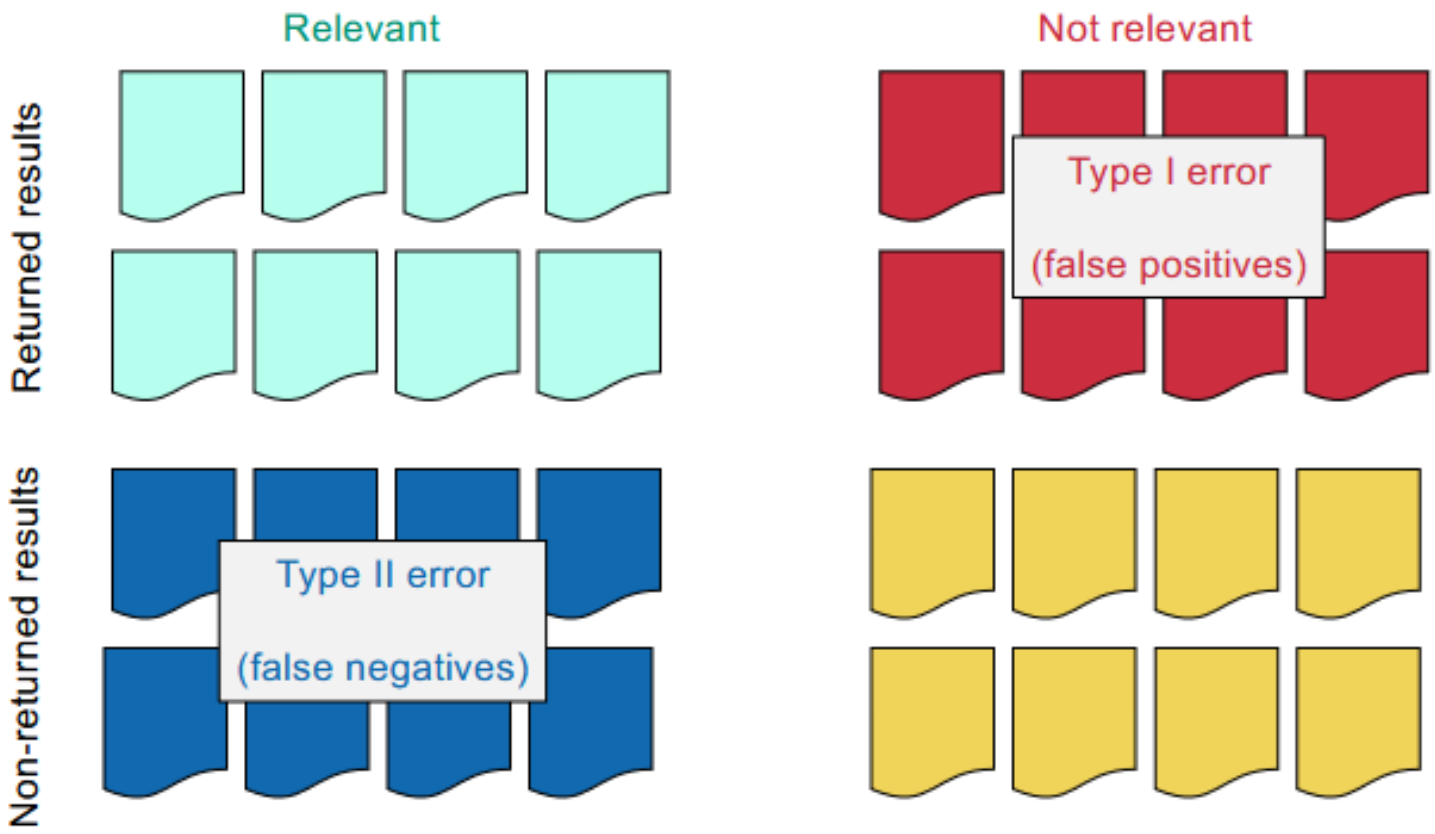
Recall: 50%

Evaluating our Results

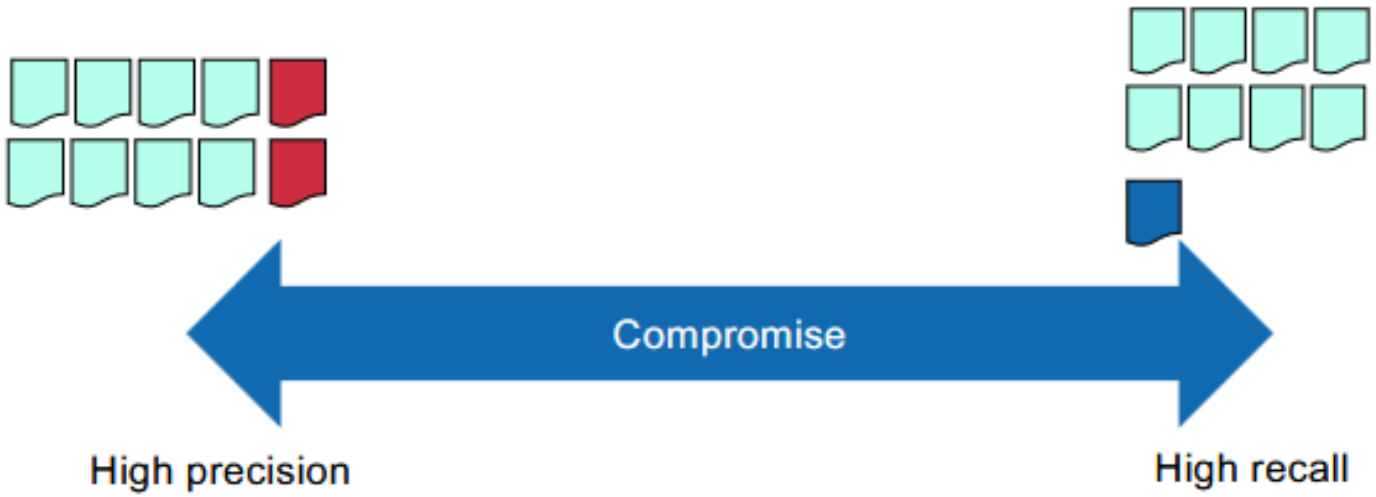


Recall: 67%

Evaluating our Results



Evaluating our Results



Exercise 1: Boolean Retrieval

- Theoretical Questions on Moodle
- Practical part on Jupyter Notebook
  - Implement simple inverted index
- Already implemented:
  - Boolean query parser (handles tokenization of documents)

Exercise 1: Boolean Retrieval

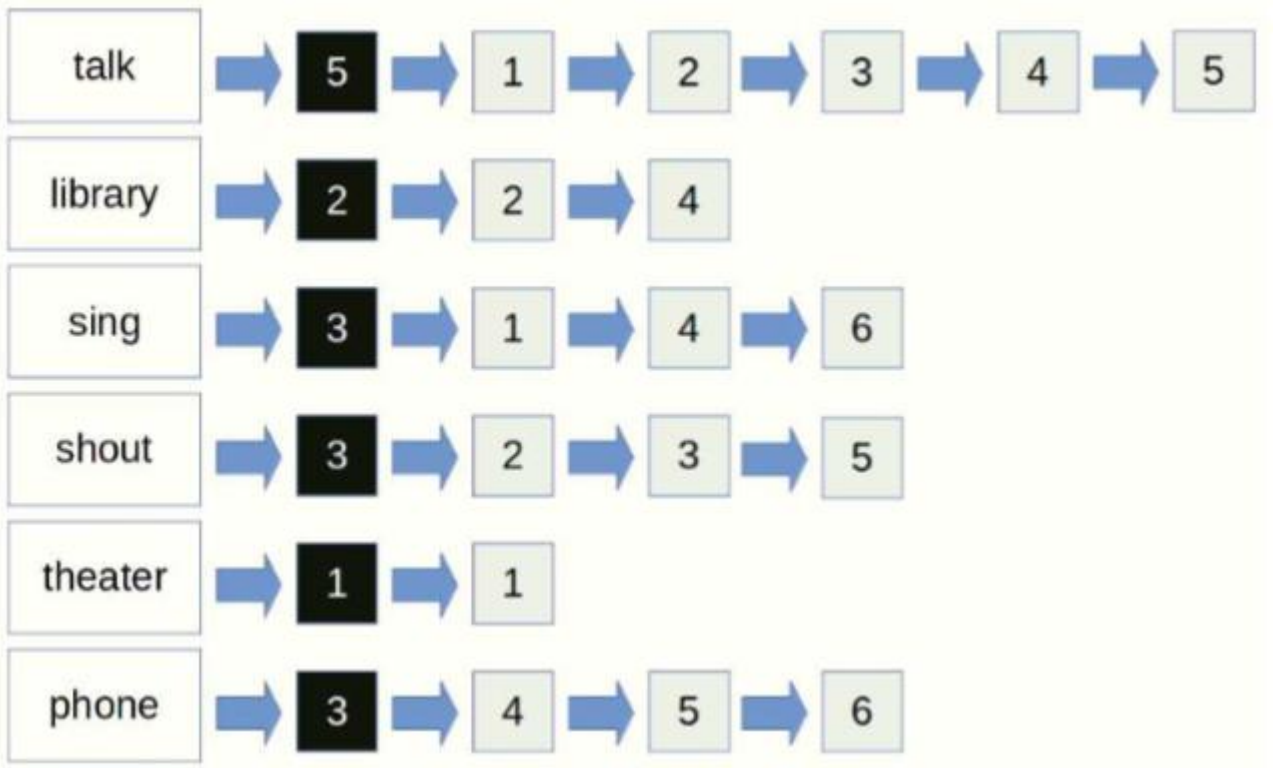
- Your part: Add a document and its terms to the inverted index

```python
paths = dict()

def add_path(path: str) -> int:
    """
    Given document path, assign unique document ID and return it
    """
    global paths # access global var we defined
    print(f"Adding {path} to path")
    # TODO 3
```

```python
index = dict()

def add_index(doc_idx: int):
    """
    Given document path and its ID, tokenize it and add the ID to the inverted in
    """
    global docs # access global var we defined
    path = paths[doc_idx]
    print(f"Adding {path} to index {doc_idx}")
    # TODO 4
```

```python
def execute_query(query: str) -> list[int]:
    """
    Retrieve the document IDs from the inverted index given query
    """
    # TODO 5
```

Exam Questions

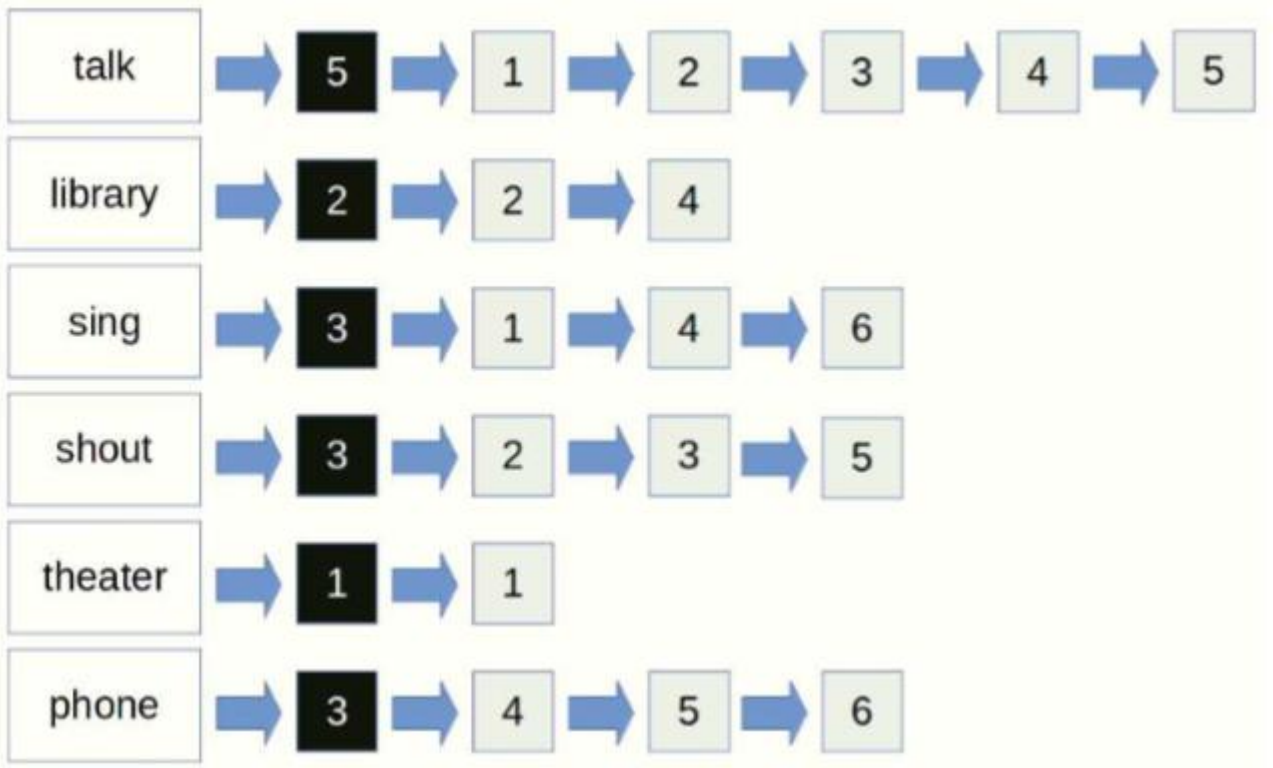


Given this inverted index, what documents does the following query return?

sing AND phone

$\{1, 4, 6\} \cap \{4, 5, 6\} = \{4, 6\}$

Exam Questions

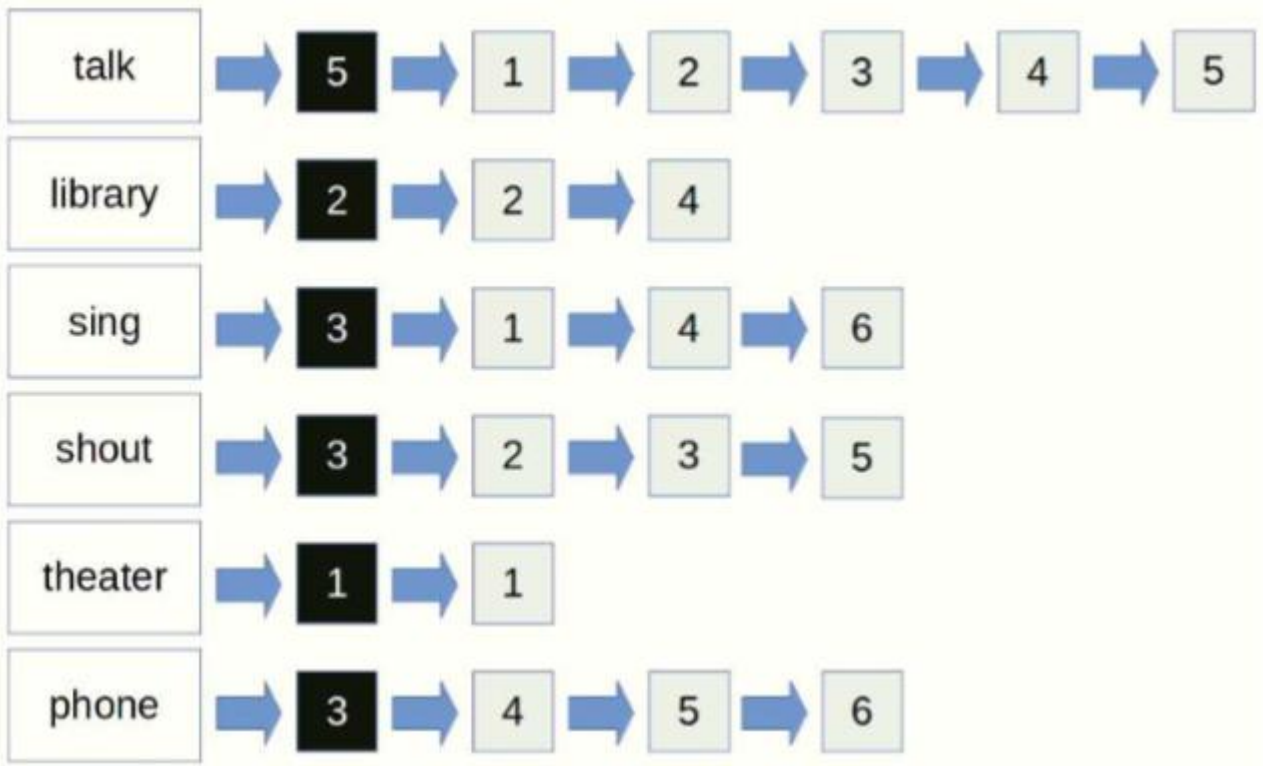


Given this inverted index, what documents does the following query return?

library AND NOT phone

$\{2, 4\} \cap \neg\{4, 5, 6\} = \{2, 4\} \cap \{1, 2, 3\} = \{2\}$

Exam Questions

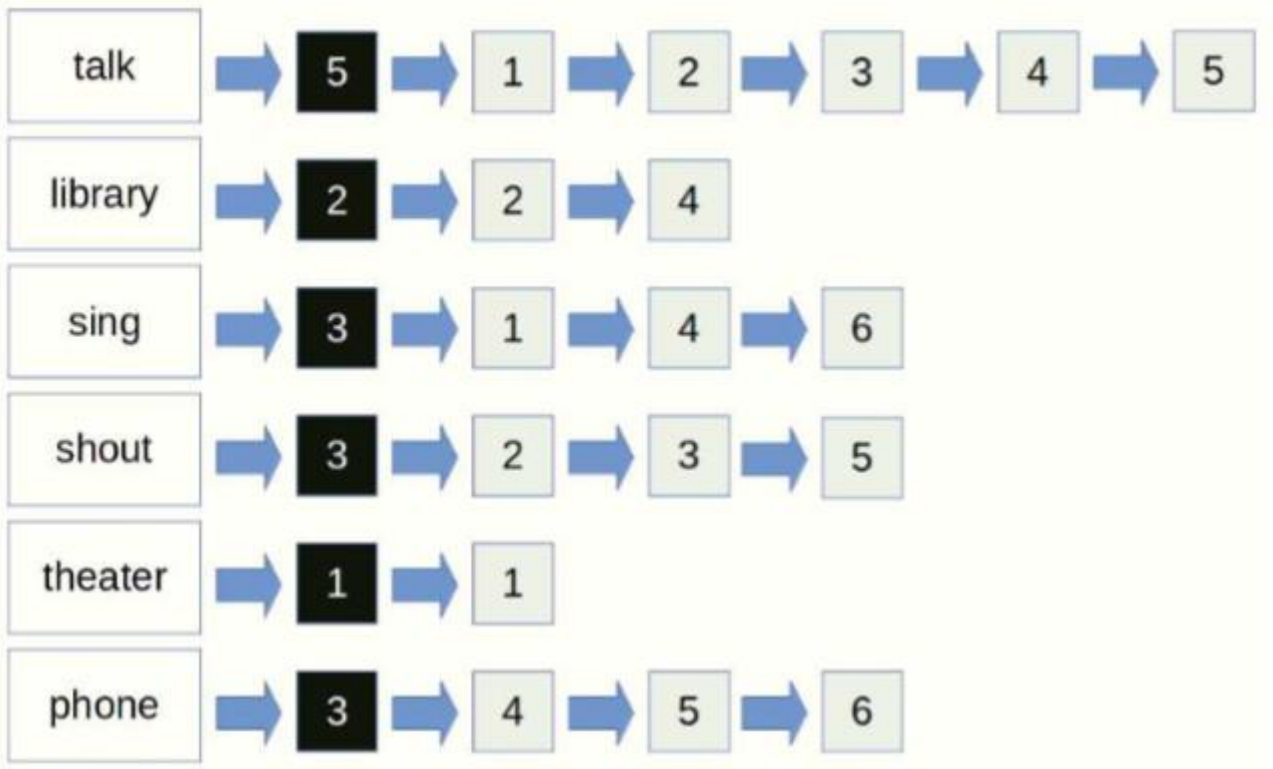


Given this inverted index, what documents does the following query return?

NOT(shout OR NOT sing)

$\neg(\{2, 3, 5\} \cup \neg\{1, 4, 6\}) = \neg(\{2, 3, 5\} \cup \{2, 3, 5\}) = \neg\{2, 3, 5\} = \{1, 4, 6\}$

Exam Questions



Given this inverted index, could the following document have been in the original collection (assuming the index shows a subset of the terms in the original collection)?

sing on the phone but not in the library

No, since «sing», «phone» and «library» only have doc 4 in common, but doc 4 also includes the term «talk», which is not in this document.

Exam Questions

We have four documents for which we want to build a standard inverted index. Each document is assigned an ID.

- **Doc 1:** new home sales top forecasts
- **Doc 2:** home sales rise in july
- **Doc 3:** increase in home sales in july
- **Doc 4:** july new home sales rise

Could the following proposal be in this standard inverted index? (incomplete postings lists are considered to be «Not in the index»)

july -> **2** -> 2 -> 4

No, since this is incomplete: july -> **3** -> 2 -> 3 -> 4

Exam Questions

We have four documents for which we want to build a standard inverted index. Each document is assigned an ID.

- **Doc 1:** new home sales top forecasts
- **Doc 2:** home sales rise in july
- **Doc 3:** increase in home sales in july
- **Doc 4:** july new home sales rise

Could the following proposal be in this standard inverted index? (incomplete postings lists are considered to be «Not in the index»)

sales -> **4** -> 1 -> 2 -> 4

No, since this is incomplete: sales -> **4** -> 1 -> 2 -> 3 -> 4

Exam Questions

We have four documents for which we want to build a standard inverted index. Each document is assigned an ID.

- **Doc 1:** new home sales top forecasts
- **Doc 2:** home sales rise in july
- **Doc 3:** increase in home sales in july
- **Doc 4:** july new home sales rise

Could the following proposal be in this standard inverted index? (incomplete postings lists are considered to be «Not in the index»)

home-> **4** -> 1 -> 2 -> 3 -> 4

Yes!

Exam Questions

We have four documents for which we want to build a standard inverted index. Each document is assigned an ID.

- **Doc 1:** new home sales top forecasts
- **Doc 2:** home sales rise in july
- **Doc 3:** increase in home sales in july
- **Doc 4:** july new home sales rise

Could the following proposal be in this standard inverted index? (incomplete postings lists are considered to be «Not in the index»)

new -> **2** -> 1 -> 4

Yes!

Exam Questions

Consider the following two postings lists:

[1, 2, 4, 8, 12, 15, 16, 32, 33, 34, 47, 81, 120, 122, 157, 180]

[15, 47]

How many comparisons would be done to intersect the two postings lists using a standard postings list structure as well as the standard intersection algorithm?

- ❑ 10
- ❑ 11
- ❑ 12
- ❑ 16

Exam Questions

Consider the following two postings lists:

[1, 2, 4, 8, 12, 15, 16, 32, 33, 34, 47, 81, 120, 122, 157, 180]

[15, 47]

How many comparisons would be done to intersect the two postings lists using a standard postings list structure as well as the standard intersection algorithm?

- ❑ 10
- ✓ 11
- ❑ 12
- ❑ 16

Kahoot

https://create.kahoot.it/share/ir-week-ii-boolean-retrieval/b04521d6-eedd-46c2-8e91-959ee741809f